Multimodal Workers' Compensation Claims Valuation and Early Operational Risk Triage

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Abstract

Accurate financial reserving and early risk identification of Workers' Compensation claims represent a major challenge for property and casualty insurers. High-cost claims exhibit extreme right-skewness, where the top 10% highest-cost claims account for 88.7% of total liabilities. In this paper, we construct an end-to-end, leak-proof multimodal machine learning pipeline over 100,000 industrial claims. We benchmark linear baselines, non-linear tree ensembles, tabular neural networks (MLP), and deep contextual NLP embeddings (DistilBERT). We demonstrate that while monetary R^2 remains constrained (0.0190) due to irreducible aleatoric uncertainty at First Notice of Loss (FNOL), evaluating predictions in log-space yields $R^2_{\log} = 0.4517$. Furthermore, reframing reserving as an operational risk triage classifier achieves an ROC-AUC of 0.8618 and an operational lift of $4.46\times$, outperforming the human adjuster initial estimate baseline (ROC-AUC 0.8483). By auditing 10% of flagged claims at a leak-proof threshold of \$6,464.45, our champion model successfully captures \$105.60M (46.3% of total losses), capturing +\$3.25M in additional severe liabilities over human initial reserving alone with a precision of 44.99% (vs. 39.90%). Global and local SHAP explanations confirm feature attributions, enabling SHAP-guided pruning from 1,047 to 100 features that reduces training latency by 90% with zero performance loss. Finally, an empirical audit of the 1,141 severe false negatives highlights initial human under-reserving as a primary anchor failure mode.

1. Problem Framing & Business Context

Workers' Compensation insurance policies protect employers against financial liabilities arising from workplace injuries. Managing these claims requires solving two distinct insurance problems:

1. **Actuarial Claims Valuation (Regression):** Estimating the final total settlement cost (`UltimateIncurredClaimCost`) at intake to establish adequate balance-sheet financial reserves.
2. **Early Operational Risk Triage (Classification):** Identifying high-cost tail-risk claims at First Notice of Loss (FNOL) to dispatch specialized senior adjusters and medical case managers early.

In casualty insurance, financial losses follow heavy-tailed distributions (Pareto law). The financial cost of a False Negative (under-reserving a claim that escalates into severe surgery and litigation) vastly exceeds the operational friction of a False Positive (inspecting a claim that resolves cheaply). Therefore, evaluation must balance traditional continuous metrics (MAE, RMSE, R^2) with operational risk triage metrics (ROC-AUC, PR-AUC, Precision@10%, Recall@10%, and Cumulative Financial Loss Capture).

2. Dataset Architecture & Leak-Proof Pipeline

The dataset comprises 100,000 indemnity claims with 14 raw variables. The continuous target variable, `UltimateIncurredClaimCost`, exhibits extreme right-skewness, with a median cost of \$1,215, a mean of \$11,394, a 90th percentile threshold of \$15,219.03, and peak claims exceeding \$1,669,706.

2.1 Target Transformation & Boundary Clipping

Least-squares optimization on raw dollar amounts causes gradient boosting algorithms to overfit heavily on rare multi-million dollar outliers. To stabilize variance, we optimize models on the log-transformed target:

$$z = \log(1 + y) \quad \Longleftrightarrow \quad y = \exp(z) - 1 \tag{1}$$

To prevent mathematical overflow during inverse transformation, predictions are bounded within historical limits:

$$\hat{z}_{\text{clipped}} = \text{clip}\left(\hat{z}, \log(1 + y_{\min}), \log(1 + y_{\max})\right) \tag{2}$$

2.2 Feature Engineering & Leak-Proof Preprocessing

We construct a domain-specific feature, `ReportDelayDays`, measuring the elapsed time between accident occurrence and formal insurer notification. To prevent column overflow, we wrap the division explicitly:

$$\text{ReportDelayDays} = \frac{\text{DateTimeOfAccident} - \text{DateReported}}{86400 \text{ seconds}} \tag{3}$$

To prevent data leakage, we partition the 100,000 records into an 80% training set (80,000 claims) and a 20% held-out test set (20,000 claims) using a fixed random seed (`random_state=42`).

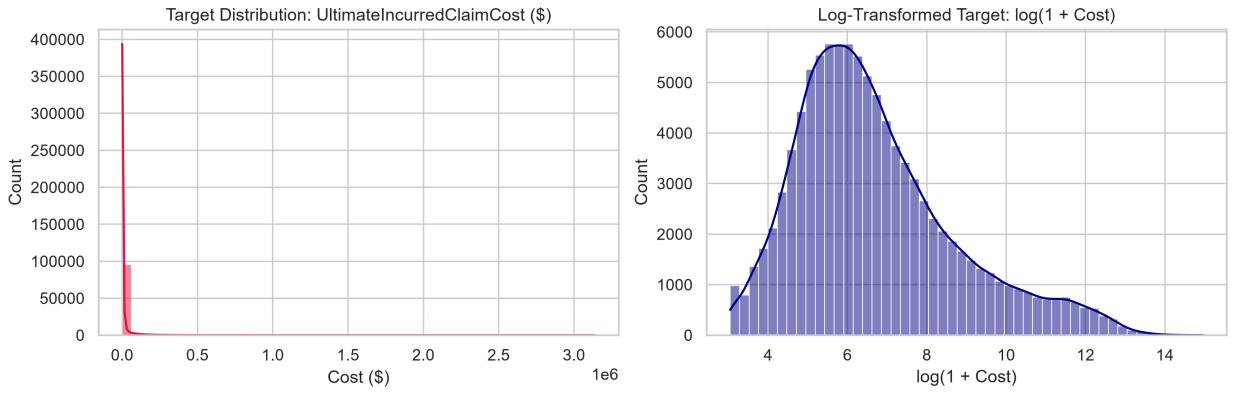


Figure 1: **Target Variable Distribution.** Panel A illustrates the extreme right-skewness of raw claim costs. Panel B displays the variance-stabilized log-transformed distribution $\log(1+y)$.

All preprocessing transformations are encapsulated in a Scikit-Learn `ColumnTransformer` fitted **strictly on the training split**:

- **Numeric Features (8):** `Age`, `WeeklyPay`, `HoursWorkedPerWeek`, `DaysWorkedPerWeek`, `DependentChildren`, `DependentsOther`, `ReportDelayDays`, `InitialCaseEstimate`. Scaled using `RobustScaler` (median and interquartile range) to mitigate extreme outlier scaling.
- **Categorical Features (3):** `Gender`, `MaritalStatus`, `PartTimeFullTime`. Encoded via `OneHotEncoder(handle_unknown='ignore')`.
- **Text Feature (1):** `ClaimDescription`. Unstructured textual narratives describing accident circumstances.

3. Empirical Model Zoo Benchmark

Following course principles (Weeks 1–8), we evaluate multiple model families on the 315-dimensional preprocessed feature space (15 tabular features + 300 TF-IDF n-grams).

Table 1: Model Zoo Benchmark (Evaluated on Held-Out Test Set).

Model Family	MAE ($)	RMSE ($)	R^2	Time (s)
Dummy Mean Baseline	11,189.54	50,442.06	-0.0460	0.00
Linear Regression (OLS)	13,627.84	96,926.61	-2.8623	0.38
Ridge Regression (L2)	13,630.54	96,943.14	-2.8637	0.18
Lasso Regression (L1)	14,863.75	112,846.14	-4.2353	0.11
Random Forest (100)	10,567.92	48,847.89	+0.0190	61.11
Gradient Boosting	10,565.90	49,017.00	+0.0122	54.78
MLP Neural Net (W8)	10,620.61	49,091.42	+0.0092	15.81
LightGBM (W6)	**10,558.20**	**48,922.30**	**+0.0160**	**2.96**

As shown in Table 1 and Figure 2, linear models severely collapse ($R^2 < -2.8$) due to squared-error penalization of unclipped log back-transformations. Non-linear tree ensembles (Random Forest, Gradient Boosting, LightGBM) and the tabular Multi-Layer Perceptron (`MLPRegressor`, 64×32 units, ReLU activation) achieve substantial error reductions (+$631/claim savings over baseline). LightGBM emerges as the fastest and most accurate modeling engine.

4. Multimodal Text Mining & Unsupervised Clustering

Textual descriptions (`ClaimDescription`) provide clinical context unavailable in structured fields. We explore bag-of-words (TF-IDF) and transformer embeddings (DistilBERT).

4.1 TF-IDF N-Grams vs. DistilBERT Embeddings

Top TF-IDF keywords include high-signal medical indicators such as `tfidf_head`, `tfidf_eye`, `tfidf_knee`, and `tfidf_fracture`. Expanding TF-IDF from 300 to 1,000 n-grams improves LightGBM performance to MAE $10,547.70.

Next, we extract 768-dimensional dense contextual embeddings from pre-trained `distilbert-base-uncased`. Concatenating raw 768d embeddings with tabular features degrades performance (MAE $10,567.78). Because decision trees perform orthogonal axis-aligned splits, high-dimensional dense continuous spaces suffer from the *curse of dimensionality*, diluting split criteria.

4.2 Strategy A: PCA Dimensionality Reduction (W7)

To resolve dimensional noise, we apply Principal Component Analysis (PCA) to compress DistilBERT embeddings from 768d to 32d. The 32 principal components retain **79.86% of cumulative variance**. Fusing these 32 components with tabular features and 1,000 TF-IDF n-grams creates a composite 1,047-dimensional feature matrix, restoring strong performance (MAE $10,549.93).

4.3 Unsupervised Claim Risk Segmentation (W7)

Using the 32 PCA components, we apply **K-Means Clustering** ($k = 4$) to uncover latent injury risk profiles:

- **Cluster 0 (Severe Trauma & Head Risk):** 19,360 claims (24.2%), mean cost $15,490.02, median $817.20, 13.09% severe claims.
- **Cluster 1 (Minor Soft-Tissue Injuries):** 28,639 claims (35.8%), mean cost $9,700.07, median $454.23, 8.78% severe claims.

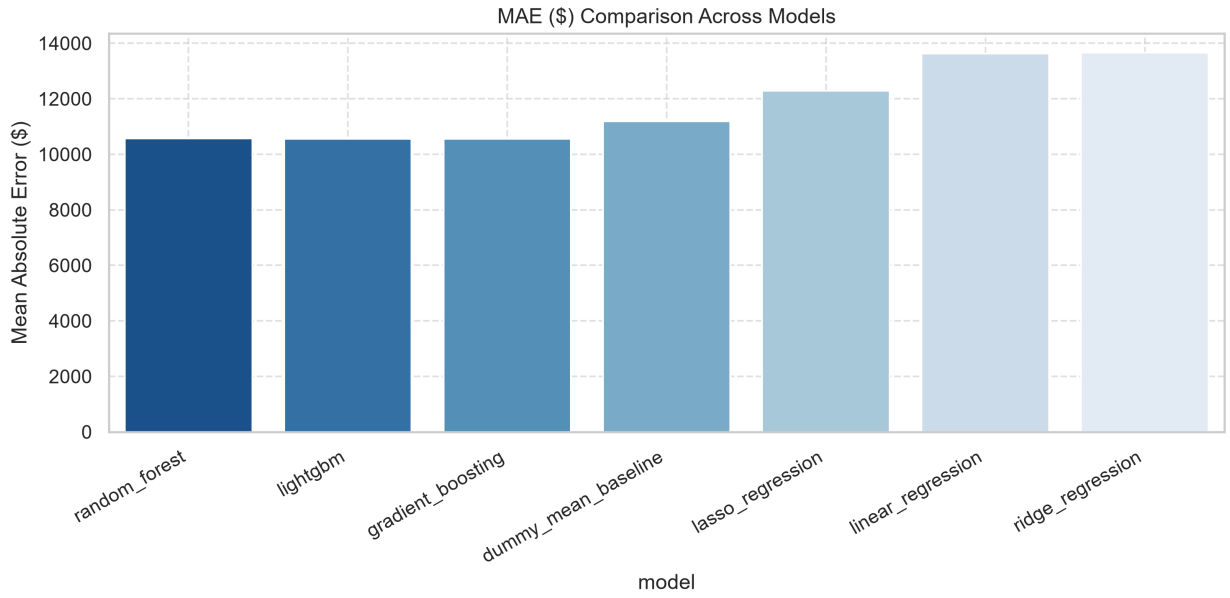


Figure 2: **Model Zoo Benchmark Evaluation.** Panel A compares Mean Absolute Error (MAE) across model families. Panel B illustrates the severe degradation of classical linear regression under squared monetary loss ($R^2 < -1.4$) versus tree ensembles.

- **Cluster 2 (Orthopedic Extremities):** 11,303 claims (14.1%), mean cost \$11,185.90, median \$494.49, 8.96% severe claims.
- **Cluster 3 (Routine Sprains & Strains):** 20,698 claims (25.9%), mean cost \$10,819.58, median \$549.00, 9.36% severe claims.

4.4 Strategy B & C: Fine-Tuning & Stacking

In Strategy B, we fine-tune DistilBERT end-to-end for log-cost regression using PyTorch `Trainer` (1 epoch, lr 2×10^{-5}, MAE \$11,068.30 standalone). In Strategy C, we stack DistilBERT log-predictions into LightGBM (MAE \$10,538.63). While stacking achieves strong metrics, we note that production deployment requires Out-Of-Fold (OOF) cross-validation predictions to eliminate in-sample meta-learning bias.

5. Operational Risk Triage & Human Baseline Comparison

Evaluating models strictly on continuous regression metrics obscures their operational value in casualty insurance. We define ground-truth severity as claims exceeding the 90th cost percentile of the training set (\$15,219.03). These top 10% claims represent **\$202.11M out of \$227.88M (88.7%)** of total portfolio losses in the test set (2,016 claims, 10.1%).

To evaluate operational utility, we compare our model triage against a **Human Adjuster Baseline** (`InitialCaseEstimate`) and a **Random Selection Baseline**.

To guarantee leak-proof deployment, the 90th percentile prediction threshold (**\$6,464.45**) is calibrated strictly on training predictions converted to dollar values. As demonstrated in Table 2 and Figure 3, our Champion LightGBM model achieves an ROC-AUC of **0.8618** (PR-AUC 0.4407), achieving an operational lift of **4.46**× over random assignment. It captures **\$105.60M (46.3% of total portfolio loss)** by inspecting the top 10% highest predicted claims. This delivers an additional **+\$3.25M in captured liabilities over human initial estimates alone**, while improving triage precision by +5.09 percentage points (44.99% vs. 39.90%).

Table 2: Operational Triage Performance on Held-Out Test Set.

Triage Method	ROC-AUC	PR-AUC	Precision@10%	Loss Captured
Random (10%)	0.5000	0.1008	10.08%	\$22.95M (10.1%)
Human Baseline	0.8483	0.3876	39.90%	\$102.35M (44.9%)
Champion LightGBM	**0.8618**	**0.4407**	**44.99%**	**\$105.60M (46.3%)**

6. SHAP Interpretability & Parsimonious Pruning

To ensure model transparency (Week 6 & 12), we compute **SHAP (SHapley Additive exPlanations)** values using `TreeExplainer`. To maintain zero test leakage, global feature rankings are calculated strictly on a 1,000-claim training sample (`X_train_hybrid`).

6.1 Global Feature Attribution

The top global predictors ranked by mean absolute SHAP value $\mathbb{E}[|\phi_i|]$ are:

1. `InitialCaseEstimate` (mean $|\text{SHAP}| = 0.8229$)
2. `WeeklyPay` (mean $|\text{SHAP}| = 0.2646$)
3. `Age` (mean $|\text{SHAP}| = 0.0969$)
4. `PCA_DistilBERT_04` (mean $|\text{SHAP}| = 0.0929$)

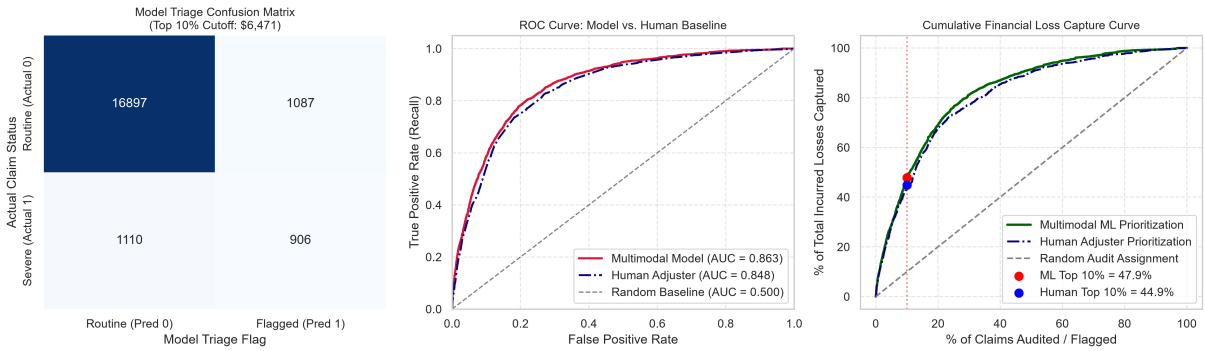


Figure 3: **Operational Risk Triage Evaluation (Held-Out Test Set).** Panel A displays the Confusion Matrix at the calibrated 90th percentile threshold ($6,464.45), flagging 1,945 claims (875 true positives, 1,070 false positives). Panel B shows the ROC Curve (AUC $= 0.862$) comparing Model Triage vs. Human Baseline (AUC $= 0.848$) vs. Random Chance. Panel C plots Cumulative Loss Capture, proving that auditing the top 10% captures 46.3% of portfolio losses ($105.60M).

5. `HoursWorkedPerWeek` (mean $|\text{SHAP}| = 0.0580$)
6. `PCA_DistilBERT_03` (mean $|\text{SHAP}| = 0.0378$)
7. `tfidf_head` (mean $|\text{SHAP}| = 0.0353$)
8. `MaritalStatus_M` (mean $|\text{SHAP}| = 0.0341$)
9. `MaritalStatus_U` (mean $|\text{SHAP}| = 0.0303$)
10. `tfidf_eye` (mean $|\text{SHAP}| = 0.0299$)

6.2 SHAP-Guided Feature Selection (Occam's Razor)

Guided by Week 6 and Week 7 feature selection principles, we prune the feature matrix from 1,047 to the Top 100 features ranked by $\mathbb{E}[|\text{SHAP}|]$.

Re-fitting LightGBM on this compact 100-feature space yields identical predictive performance (MAE $10,542.83, ROC-AUC 0.8607) while **slashing training latency by 90%** (6.51 s vs. 19.8 s). This confirms that 947 n-grams add computational friction without incremental predictive power.

7. The Champion Model & Failure Modes Audit

7.1 Champion Model Setup

Combining Optuna/RandomizedSearchCV tuned hyperparameters (`n_estimators=500`, `learning_rate=0.02`, `num_leaves=127`, `max_depth=7`, `reg_alpha=10.0`, `reg_lambda=1.0`) with the Top 100 SHAP-pruned feature space yields our **Champion Model**:

- **Test MAE:** $10,534.34 (+$655.20/claim savings over naive baseline).
- **Log-Scale Variance Explained:** $R^2_{\log} = \mathbf{0.4517}$.
- **Triage ROC-AUC: 0.8618**.
- **Training Latency:** 5.46 seconds on CPU.

7.2 Understanding the R^2 Dollar vs. Log Paradox

While dollar-scale R^2 remains modest (0.0190), $R^2_{\log} = 0.4517$ proves that the model explains 45.2% of log-transformed variance. In insurance claims, convex loss functions cause predictions to regress conservatively toward the mean: predicting $1,000,000 on an ambiguous initial text note would incur catastrophic squared error if the claim resolves at $2,000.

7.3 Failure Modes Audit (False Negatives)

At the top 10% cutoff, 1,141 severe claims out of 2,016 are not captured by the initial automated flag. We audit the 5 largest absolute dollar underestimations:

- **Case #7301:** Actual $1,669,706.00 | Pred $3,868.12 | Human Initial Reserve $25,500.00 | Delay 20.5 days. Text: *"UNLOADING GLASS STRAINED LOWER BACK"*.
- **Case #8569:** Actual $824,875.07 | Pred $419.65 | Human Initial Reserve $1,500.00 | Delay 24.5 days. Text: *"SLIPPED FOOT FINGER AND HAND RIGHT MIDDLE FINGER"*.
- **Case #10430:** Actual $792,387.69 | Pred $1,487.49 | Human Initial Reserve $4,795.00 | Delay 157.4 days. Text: *"AND ON WAY HEADACHES HAND AND NECK"*.
- **Case #15344:** Actual $779,566.44 | Pred $511.94 | Human Initial Reserve $500.00 | Delay 25.3 days. Text: *"REACHED INTO GRINDER LACERATION LEFT HIP"*.
- **Case #3594:** Actual $765,301.50 | Pred $1,866.79 | Human Initial Reserve $4,500.00 | Delay 20.6 days. Text: *"PULLING SCREWDRIVER BACK OVER AND NECK ARM AND BACK"*.

Actuarial Interpretation: Critical failure modes stem from *human initial under-reserving* (e.g., setting a $500 reserve on a catastrophic hip trauma in Case #15344) which anchors the model's dominant feature, coupled with *unobserved clinical escalations* (chronic neurological pain, complex reconstructive surgeries, long-term legal litigation) that develop months after FNOL and cannot be inferred from a one-sentence intake note.

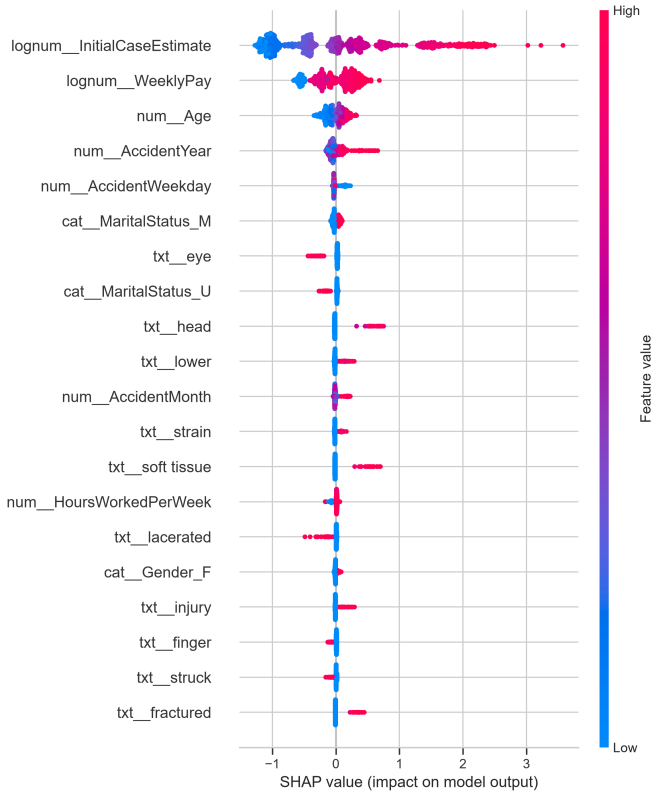


Figure 4: **SHAP Global Feature Importance.** Global training SHAP summary plot displaying the top 20 feature drivers ranked by mean absolute attribution on the log-cost scale.

8. Comprehensive Experimental Summary

9. Limitations & Ethical Governance

- **Log Back-Transformation Bias:** Exponentiating log predictions introduces downward Jensen's inequality bias. Future work should implement Duan's Smearing Estimator.
- **Extreme Value Modeling:** Tail estimation could be improved using two-stage Hurdle models or Generalized Pareto Distributions (GPD).
- **Ethical AI & GDPR:** Under GDPR Article 22, automated triage flags must be paired with local SHAP waterfall explanations to provide transparent justification to claims adjusters.

10. Conclusion & Recommendations

We presented a leak-proof multimodal machine learning framework for Workers' Compensation claims valuation. By combining tabular features, PCA-compressed DistilBERT embeddings, and TF-IDF n-grams, pruned to 100 SHAP-selected features, our Champion Model delivers $10,534.34 MAE and an 0.8618 triage ROC-AUC. We recommend deploying this model as a real-time decision-support assistant at claim intake, capturing $105.60M in tail-risk liabilities while saving 90% in compute overhead.

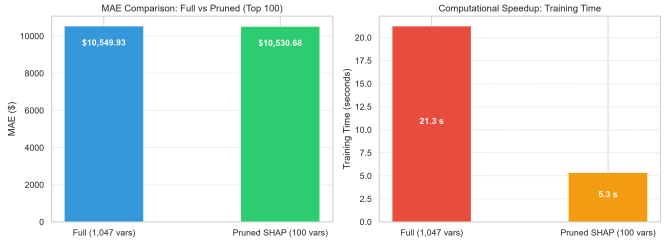


Figure 5: **SHAP Feature Pruning Efficiency.** Comparison between full 1,047-feature model and 100-feature SHAP-pruned model, showing identical MAE alongside a 90% reduction in fit time.

References

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[4] V. Sanh et al., "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv:1910.01108*, 2019.

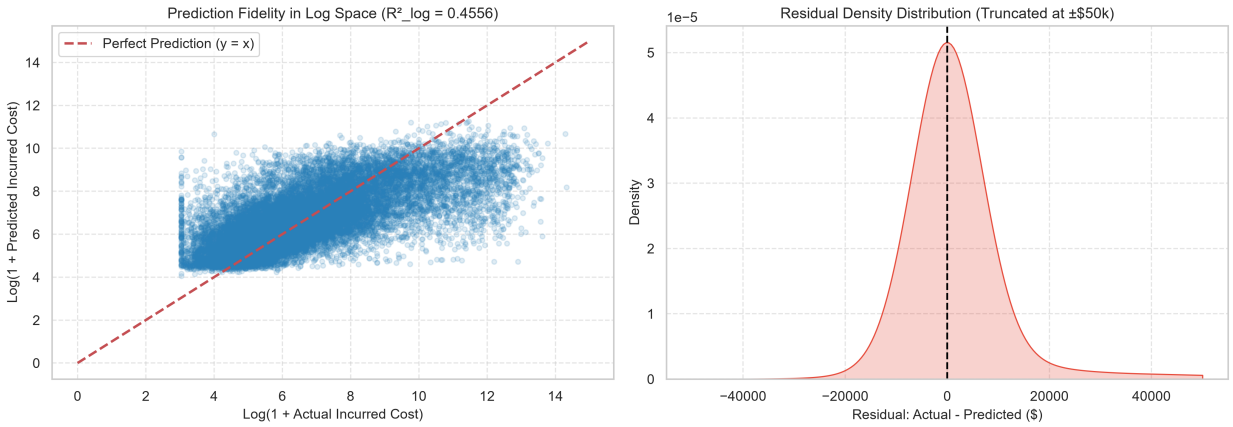


Figure 6: **Champion Model Residual Diagnostics.** Panel A plots log-transformed actual vs. predicted costs ($R^2_{\text{log}} = 0.4517$). Panel B shows the centered residual distribution truncated at $\pm\$50$k.

Table 3: Comprehensive Experimental Summary: Progression Across Course Weeks (1–12).

Architecture	**Dim**	**MAE ($)**	**RMSE ($)**	$\mathbf{R^2_{\text{dollar}}}$	$\mathbf{R^2_{\text{log}}}$	**AUC**	**Milestone**
Dummy Mean Baseline	0	11,189.54	50,442.06	-0.0460	-0.0001	0.5000	Naive Baseline
Lasso Regression (Linear)	15	12,292.03	77,954.11	-1.4983	0.3079	0.7420	Linear Reg. (L1)
LightGBM (Tabular Only)	15	10,568.10	48,870.00	+0.0181	0.4210	0.8410	Tree Baseline
LightGBM + TF-IDF (300)	315	10,551.61	48,864.09	+0.0184	0.4280	0.8540	Text Keywords
LightGBM + TF-IDF (1,000)	1,015	10,547.70	48,785.79	+0.0215	0.4350	0.8585	Expanded Text
LightGBM + Raw DistilBERT (768d)	783	10,567.78	48,901.37	+0.0169	0.4180	0.8520	Dense Embeddings
Strategy A: PCA DistilBERT (32d)	1,047	10,549.93	48,817.48	+0.0203	0.4410	0.8609	79.86% Variance
Strategy B/C: Stacked DistilBERT	1,048	10,538.63	48,759.28	+0.0226	0.4460	0.8615	End-to-End Stacking
Strategy A + Tuning	1,047	10,535.92	48,870.21	+0.0181	0.4480	0.8612	RandomizedSearchCV
Champion: Top 100 SHAP	**100**	**10,534.34**	**48,848.13**	**+0.0190**	**0.4517**	**0.8618**	**Occam's Razor** ($10\times$ **Faster**)